

Course 3062

3 Credits: 3

Program core

Source-grounded

Textile Wet Processing

□ Identify the chemicals used and pre-treatments given to cotton fibres.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 3062 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3062.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Textile Technology
Course code	3062
Course title	Textile Wet Processing
Semester	3 Credits: 3
Course category	Program core
Periods per week	3 (L:2, T:1, P:0) Periods per semester: 45
Periods per semester	45

Item	Official information
Credits	3
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

12 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- Identify the chemicals used and pre-treatments given to cotton fibres.
- To impart the knowledge of dyeing natural cellulosic & Viscose fibres using water-soluble & water-insoluble dyes.

Course outcomes

CO	Official outcome	Study level
CO1	11 Understanding fabrics Discuss various methods of bleaching textile	See official syllabus
CO2	materials & Cotton Mercerisation process. 10 Understanding and Mercerizing Discuss the process of dyeing Cotton/Viscose fibre	See official syllabus
CO3	10 Understanding using water-soluble dyes. Explain the process of cotton/ viscose fibre dyeing	See official syllabus
CO4	12 Understanding with water insoluble dyes.	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Demonstrate the pre-treatments given to Cotton Grey fabrics	3	As specified
Module 2	Use of various methods of bleaching textile materials & Cotton	2	As specified
Module 3	dyes.	4	As specified
Module 4	Execute cotton/ viscose fibre dyeing with water insoluble dyes.	3	As specified

MODULE 1

Demonstrate the pre-treatments given to Cotton Grey fabrics

This module is presented from the official syllabus scope for Textile Wet Processing. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Demonstrate the pre-treatments given to Cotton Grey fabrics

- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

know various chemicals used in textile wet 4 Remembering processing

Discuss the preparation of cotton/ Viscose for

know various chemicals used in textile wet 4 Remembering processing Discuss the preparation of cotton/ Viscose for is an official topic in Module 1 of Textile Wet Processing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify know various chemicals used in textile wet 4 Remembering processing Discuss the preparation of cotton/ Viscose for using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, know various chemicals used in textile wet 4 remembering processing discuss the preparation of cotton/ viscose for should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What know various chemicals used in textile wet 4 Remembering processing Discuss the preparation of cotton/ Viscose for means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പ്രതിപദം വിവിധ രാസവസ്തുക്കൾ ഉപയോഗിച്ചുള്ള വസ്തുനിർമ്മാണത്തിന്റെ തത്വം ഓർമ്മപ്പെടുത്തുക. വിവിധ രാസവസ്തുക്കൾ ഉപയോഗിച്ചുള്ള വസ്തുനിർമ്മാണത്തിന്റെ തത്വം ഓർമ്മപ്പെടുത്തുക. Remembering processing Discuss the preparation of cotton/ Viscose for പദാർത്ഥത്തിന്റെ നിർമ്മാണത്തെക്കുറിച്ച് ചർച്ച ചെയ്യുക. നിർമ്മാണത്തിന്റെ തത്വം ഓർമ്മപ്പെടുത്തുക, steps, application പദാർത്ഥത്തിന്റെ നിർമ്മാണത്തെക്കുറിച്ച് ചർച്ച ചെയ്യുക. Technical terms English-ൽ പദാർത്ഥത്തിന്റെ നിർമ്മാണത്തെക്കുറിച്ച് ചർച്ച ചെയ്യുക.

4 Understanding dyeing

4 Understanding dyeing is an official topic in Module 1 of Textile Wet Processing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding dyeing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding dyeing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding dyeing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-4 Understanding dyeing ധാരാളം ധാരാളം definition/meaning ധാരാളം. ധാരാളം ധാരാളം, steps, application ധാരാളം ധാരാളം. Technical terms English-4 ധാരാളം ധാരാളം.

Differentiate Singeing, Desizing & Scouring 3 Understanding Chemistry of Wetting: Definition of surface tension and contact angle - Relationship of surface tension and contact angle with wetting of textile materials Preparation of Cotton/ Viscose for Dyeing: Sequence of operations in wet processing of Cotton/ Viscose material. Singeing: Objectives - Methods - Hot plate, Roller, Gas singeing machines. De-sizing: Objectives - Methods - Rot steeping - Acid steeping, Enzymatic de-sizing. Scouring: Objectives and importance - mechanism of scouring- Types of Kiers. Features of a modern Kiers and its working

Differentiate Singeing, Desizing & Scouring 3 Understanding Chemistry of Wetting: Definition of surface tension and contact angle - Relationship of surface tension and contact angle with wetting of textile materials Preparation of Cotton/ Viscose for Dyeing: Sequence of operations in wet processing of Cotton/ Viscose material. Singeing: Objectives - Methods - Hot plate, Roller, Gas singeing machines. De-sizing: Objectives - Methods - Rot steeping - Acid steeping, Enzymatic de-sizing. Scouring: Objectives and importance - mechanism of scouring- Types of Kiers. Features of a modern Kiers and its working is an official topic in Module 1 of Textile Wet Processing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Differentiate Singeing, Desizing & Scouring 3 Understanding Chemistry of Wetting: Definition of surface tension and contact angle - Relationship of surface tension and contact angle with wetting of textile materials Preparation of Cotton/ Viscose for Dyeing: Sequence of operations in wet processing of Cotton/ Viscose material. Singeing: Objectives - Methods - Hot plate, Roller, Gas singeing machines. De-sizing: Objectives - Methods - Rot steeping - Acid steeping, Enzymatic de-sizing. Scouring: Objectives and importance - mechanism of scouring- Types of Kiers. Features of a modern Kiers and its working using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

Malayalam support: Module 1-
Understanding Chemistry of Wetting: Definition of surface tension and contact angle - Relationship of surface tension and contact angle with wetting of textile materials Preparation of Cotton/ Viscose for Dyeing: Sequence of operations in wet processing of Cotton/ Viscose material. Singeing: Objectives - Methods - Hot plate, Roller, Gas singeing machines. De-sizing: Objectives - Methods - Rot steeping - Acid steeping, Enzymatic de-sizing. Scouring: Objectives and importance - mechanism of scouring- Types of Kiers. Features of a modern Kiers and its working
definition/meaning.

steps, application ഏകദേശം അനുസരിച്ച് തിരഞ്ഞെടുക്കുക. Technical terms English-ൽ അനുസരിച്ച് തിരഞ്ഞെടുക്കുക.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Demonstrate the pre-treatments given to Cotton Grey fabrics

M1.01	Memorize the chemistry of Wetting and know various chemicals used in textile wet processing	4
M1.02	Discuss the preparation of cotton/ Viscose for dyeing	4
M1.03	Differentiate Singeing, Desizing & Scouring	3

Contents:

Chemistry of Wetting: Definition of surface tension and contact angle -

Relationship of

surface tension and contact angle with wetting of textile materials

Preparation of Cotton/ Viscose for Dyeing: Sequence of operations in wet processing of

Cotton/ Viscose material.

Singeing: Objectives - Methods - Hot plate, Roller, Gas singeing machines.

De-sizing: Objectives - Methods - Rot steeping - Acid steeping, Enzymatic de-sizing.

Scouring: Objectives and importance - mechanism of scouring- Types of Kiers. Features

of a modern Kiers and its working

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

Use of various methods of bleaching textile materials & Cotton

This module is presented from the official syllabus scope for Textile Wet Processing. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Use of various methods of bleaching textile materials & Cotton
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

2 Understanding materials Use of continuous method of scouring and Bleaching process & compare hypochlorite 4 Understanding M2.02 and peroxide bleaching Gain the knowledge on the process and

2 Understanding materials Use of continuous method of scouring and Bleaching process & compare hypochlorite 4 Understanding M2.02 and peroxide bleaching Gain the knowledge on the process and is an official topic in Module 2 of Textile Wet Processing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding materials Use of continuous method of scouring and Bleaching process & compare hypochlorite 4 Understanding M2.02 and peroxide bleaching Gain the knowledge on the process and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding materials use of continuous method of scouring and bleaching process & compare hypochlorite 4 understanding m2.02 and peroxide bleaching gain the knowledge on the process and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding materials Use of continuous method of scouring and Bleaching process & compare hypochlorite 4 Understanding M2.02 and peroxide bleaching Gain the knowledge on the process and means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 2 Understanding materials Use of continuous method of scouring and Bleaching process & compare hypochlorite 4 Understanding M2.02 and peroxide bleaching Gain the knowledge on the process and definition/meaning. , steps, application . Technical terms English-

importance of Mercerisation to implement the 4 Understanding same in new situation. Bleaching: Objectives, Types of bleaching agents-Comparison of sodium and calcium hypochlorite's in terms of their properties, chemical behaviour and application - Method of bleaching with Sodium hypochlorite. Properties of hydrogen peroxide- Strength of hydrogen peroxide in volume & percentage - Bleaching with Hydrogen peroxide (Batch process).Continuous scouring and bleaching by using 'J' Box. Compare hypochlorite and peroxide bleaching. Mercerisation: Objects of mercerizing. Changes in the appearance of cotton by mercerizing - Various factors affecting the process of Mercerisation. Working of a Hank Mercerizing machine. Working of continuous Cloth Mercerizing machine Define BAN. Implement the process of dyeing Cotton/Viscose fibre using water-soluble

importance of Mercerisation to implement the 4 Understanding same in new situation. Bleaching: Objectives, Types of bleaching agents-Comparison of sodium and calcium hypochlorite's in terms of their properties, chemical behaviour and application - Method of bleaching with Sodium hypochlorite. Properties of hydrogen peroxide- Strength of hydrogen peroxide in volume & percentage - Bleaching with Hydrogen peroxide (Batch process).Continuous scouring and bleaching by using 'J' Box. Compare hypochlorite and peroxide bleaching. Mercerisation: Objects of mercerizing. Changes in the appearance of cotton by mercerizing - Various factors affecting the process of Mercerisation. Working of a Hank Mercerizing machine. Working of continuous Cloth Mercerizing machine Define BAN. Implement the process of dyeing Cotton/Viscose fibre using water-soluble is an official topic in Module 2 of Textile Wet Processing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify importance of Mercerisation to implement the 4 Understanding same in new situation. Bleaching: Objectives, Types of bleaching agents- Comparison of sodium and calcium hypochlorite's in terms of their properties, chemical behaviour and application - Method of bleaching with Sodium hypochlorite. Properties of hydrogen peroxide- Strength of hydrogen peroxide in volume & percentage - Bleaching with Hydrogen peroxide (Batch

process).Continuous scouring and bleaching by using 'J' Box. Compare hypochlorite and peroxide bleaching. Mercerisation: Objects of mercerizing. Changes in the appearance of cotton by mercerizing - Various factors affecting the process of Mercerisation. Working of a Hank Mercerizing machine. Working of continuous Cloth Mercerizing machine Define BAN. Implement the process of dyeing Cotton/Viscose fibre using water-soluble using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, importance of mercerisation to implement the 4 understanding same in new situation. bleaching: objectives, types of bleaching agents-comparison of sodium and calcium hypochlorite's in terms of their properties, chemical behaviour and application - method of bleaching with sodium hypochlorite. properties of hydrogen peroxide- strength of hydrogen peroxide in volume & percentage - bleaching with hydrogen peroxide (batch process).continuous scouring and bleaching by using 'j' box. compare hypochlorite and peroxide bleaching. mercerisation: objects of mercerizing. changes in the appearance of cotton by mercerizing - various factors affecting the process of mercerisation. working of a hank mercerizing machine. working of continuous cloth mercerizing machine define ban. implement the process of dyeing cotton/viscose fibre using water-soluble should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What importance of Mercerisation to implement the 4 Understanding same in new situation. Bleaching: Objectives, Types of bleaching agents-Comparison of sodium and calcium hypochlorite's in terms of their properties, chemical behaviour and application - Method of bleaching with Sodium hypochlorite. Properties of hydrogen peroxide- Strength of hydrogen peroxide in volume & percentage - Bleaching with Hydrogen peroxide (Batch process).Continuous scouring and bleaching by using 'J' Box. Compare hypochlorite and peroxide bleaching. Mercerisation: Objects of mercerizing. Changes in the appearance of cotton by mercerizing - Various factors affecting the process of Mercerisation. Working of a Hank Mercerizing machine. Working of continuous Cloth Mercerizing machine Define BAN. Implement the process of dyeing Cotton/Viscose fibre using water-soluble means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഈ importance of Mercerisation to implement the 4 Understanding same in new situation. Bleaching: Objectives, Types of bleaching agents-Comparison of sodium and calcium hypochlorite's in terms of their properties, chemical behaviour and application - Method of bleaching with Sodium hypochlorite. Properties of hydrogen peroxide- Strength of hydrogen peroxide in volume & percentage - Bleaching with Hydrogen peroxide (Batch process).Continuous scouring and bleaching by using 'J' Box. Compare hypochlorite and peroxide bleaching. Mercerisation: Objects of mercerizing. Changes in the appearance of cotton by mercerizing - Various factors affecting the process of Mercerisation. Working of a Hank Mercerizing machine. Working of continuous Cloth Mercerizing machine Define BAN. Implement the process of dyeing Cotton/Viscose fibre using water-soluble ഓക്സിജൻ ഓക്സൈഡ് definition/meaning ഓക്സിജൻ ഓക്സൈഡ്. ഓക്സിജൻ ഓക്സൈഡ് ഓക്സൈഡ്, steps, application ഓക്സിജൻ ഓക്സൈഡ് ഓക്സൈഡ്. Technical terms English-ഈ ഓക്സിജൻ ഓക്സൈഡ്.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Use of various methods of bleaching textile materials & Cotton

Mercerisation process.

Classify the bleaching process given to cotton

M2.01

2

Understanding

materials

Use of continuous method of scouring and

Bleaching process & compare hypochlorite

4

Understanding

M2.02

and peroxide bleaching

Gain the knowledge on the process and

M2.03

importance of Mercerisation to implement the

4

Understanding

same in new situation.

Series Test - I

1

Contents:

Bleaching: Objectives, Types of bleaching agents-Comparison of sodium and calcium hypochlorite's in terms of their properties, chemical behaviour and application - Method of

bleaching with Sodium hypochlorite.

Properties of hydrogen peroxide- Strength of hydrogen peroxide in volume & percentage -

Bleaching with Hydrogen peroxide (Batch process).Continuous scouring and bleaching by

using 'J' Box. Compare hypochlorite and peroxide bleaching.

Mercerisation: Objects of mercerizing. Changes in the appearance of cotton by

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

dyes.

This module is presented from the official syllabus scope for Textile Wet Processing. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: dyes.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

2 Understanding application Execute the application of Direct dyes,

2 Understanding application Execute the application of Direct dyes, is an official topic in Module 3 of Textile Wet Processing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding application Execute the application of Direct dyes, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding application execute the application of direct dyes, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding application Execute the application of Direct dyes, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3- 2 Understanding application Execute the application of Direct dyes, definition/meaning .
 , steps, application . Technical terms English- .

3 Understanding Reactive dyes on Cotton/ Viscose fibres Execute the application of Basic dyes on

3 Understanding Reactive dyes on Cotton/ Viscose fibres Execute the application of Basic dyes on is an official topic in Module 3 of Textile Wet Processing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Reactive dyes on Cotton/ Viscose fibres Execute the application of Basic dyes on using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding reactive dyes on cotton/ viscose fibres execute the application of basic dyes on should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding Reactive dyes on Cotton/ Viscose fibres Execute the application of Basic dyes on means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-3 Understanding Reactive dyes on Cotton/Viscose fibres Execute the application of Basic dyes on $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^-$ definition/meaning $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^-$. $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^-$ $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^-$, steps, application $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^-$ $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^-$. Technical terms English- $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^-$ $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^-$.

3 Understanding Cotton material

3 Understanding Cotton material is an official topic in Module 3 of Textile Wet Processing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Cotton material using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding cotton material should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding Cotton material means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-3 Understanding Cotton material
 definition/meaning .
 application
 Technical terms English-
 .

Test the fastness properties of dyed material 2 Understanding Introduction to Dyeing: Define dyes - selection of dye - Classification of dyes based on the method of application - Terminologies related to dyes and dyeing - Material Liquor ratio, Shade Percentage, Percentage of Exhaustion, affinity, substantivity. General mechanism of dyeing. Important Natural dyes. Direct Dyes: Method of application of Direct Dye on Cotton and Viscose. After - treatments given to Direct dyed materials. Reactive Dyes: Properties - Types - method of application of Reactive Dyes on Cotton & Viscose Method of application of Basic dyes on cotton. Fastness properties of dyed material

Test the fastness properties of dyed material 2 Understanding Introduction to Dyeing: Define dyes - selection of dye - Classification of dyes based on the method of application - Terminologies related to dyes and dyeing - Material Liquor ratio, Shade Percentage, Percentage of Exhaustion, affinity, substantivity. General mechanism of dyeing. Important Natural dyes. Direct Dyes: Method of application of Direct Dye on Cotton and Viscose. After - treatments given to Direct dyed materials. Reactive Dyes: Properties - Types - method of application of Reactive Dyes on Cotton & Viscose Method of application of Basic dyes on cotton. Fastness properties of dyed material is an official topic in Module 3 of Textile Wet Processing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Test the fastness properties of dyed material 2 Understanding Introduction to Dyeing: Define dyes - selection of dye - Classification of dyes based on the method of application - Terminologies related to dyes and dyeing - Material Liquor ratio, Shade Percentage, Percentage of Exhaustion, affinity, substantivity. General mechanism of dyeing. Important Natural dyes. Direct Dyes: Method of application of Direct Dye on Cotton and Viscose. After - treatments given to Direct dyed materials. Reactive Dyes: Properties - Types - method of application of Reactive Dyes on Cotton & Viscose Method of application of Basic dyes on cotton. Fastness properties of dyed material using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, test the fastness properties of dyed material 2 understanding introduction to dyeing: define dyes - selection of dye - classification of dyes based on the method of application - terminologies related to dyes and dyeing - material liquor ratio, shade percentage, percentage of exhaustion, affinity, substantivity. general mechanism of dyeing. important natural dyes. direct dyes: method of application of direct dye on cotton and viscose. after - treatments given to direct dyed materials. reactive dyes: properties - types - method of application of reactive dyes on cotton & viscose method of application of basic dyes on cotton. fastness properties of dyed material should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Test the fastness properties of dyed material 2 Understanding Introduction to Dyeing: Define dyes - selection of dye - Classification of dyes based on the method of application - Terminologies related to dyes and dyeing - Material Liquor ratio, Shade Percentage, Percentage of Exhaustion, affinity, substantivity. General mechanism of dyeing. Important Natural dyes. Direct Dyes: Method of application of Direct Dye on Cotton and Viscose. After - treatments given to Direct dyed materials. Reactive Dyes: Properties - Types - method of application of Reactive Dyes on Cotton & Viscose Method of application of Basic dyes on cotton. Fastness properties of dyed material means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Test the fastness properties of dyed material 2 Understanding Introduction to Dyeing: Define dyes - selection of dye - Classification of dyes based on the method of application - Terminologies related to dyes and dyeing - Material Liquor ratio, Shade Percentage, Percentage of Exhaustion, affinity, substantivity. General mechanism of dyeing. Important Natural dyes. Direct Dyes: Method of application of Direct Dye on Cotton and Viscose. After - treatments given to Direct dyed materials. Reactive Dyes: Properties - Types - method of application of Reactive Dyes on Cotton & Viscose Method of application of Basic dyes on

cotton. Fastness properties of dyed material രണ്ടാം പാഠ്യപുസ്തകം
 definition/meaning പദാർത്ഥം. പദാർത്ഥം പദാർത്ഥം, steps, application
പദാർത്ഥം പദാർത്ഥം പദാർത്ഥം. Technical terms English-പദാർത്ഥം പദാർത്ഥം.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
 Educational explanations below are separate elaboration.

dyes.

	Classify dyes based on the method of	
M3.01		2
Understanding	application	
	Execute the application of Direct dyes,	
M3.02		3
Understanding	Reactive dyes on Cotton/ Viscose fibres	
	Execute the application of Basic dyes on	
M3.03		3
Understanding	Cotton material	
M3.04	Test the fastness properties of dyed material	2
Understanding		

Contents:

Introduction to Dyeing: Define dyes - selection of dye - Classification of dyes based on the method of application - Terminologies related to dyes and dyeing - Material Liquor ratio, Shade Percentage, Percentage of Exhaustion, affinity, substantivity. General mechanism of dyeing. Important Natural dyes. Direct Dyes: Method of application of Direct Dye on Cotton and Viscose. After -

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

Execute cotton/ viscose fibre dyeing with water insoluble dyes.

This module is presented from the official syllabus scope for Textile Wet Processing. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Execute cotton/ viscose fibre dyeing with water insoluble dyes.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

4 Remembering viscose materials with Water-insoluble dyes Application of Ingrain dyes on textile

4 Remembering viscose materials with Water-insoluble dyes Application of Ingrain dyes on textile is an official topic in Module 4 of Textile Wet Processing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Remembering viscose materials with Water-insoluble dyes Application of Ingrain dyes on textile using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 remembering viscose materials with water-insoluble dyes application of ingrain dyes on textile should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Remembering viscose materials with Water-insoluble dyes Application of Ingrain dyes on textile means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-പാഠ്യ 4 Remembering viscose materials with Water-insoluble dyes Application of Ingrain dyes on textile പദപരിചയം പദപരിചയം definition/meaning പദപരിചയം പദപരിചയം. പദപരിചയം പദപരിചയം പദപരിചയം, steps, application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms English-പദപരിചയം പദപരിചയം പദപരിചയം.

4 Understanding materials

4 Understanding materials is an official topic in Module 4 of Textile Wet Processing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding materials using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding materials should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding materials means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 4 Understanding materials ഓരോന്നിനും നിർവ്വചനം/അർത്ഥം നിർവ്വചിക്കുക. ഓരോന്നിന്റെ ഉപയോഗം, steps, application ഓരോന്നിന്റെയും ഉപയോഗം നിർവ്വചിക്കുക. Technical terms English- 4 ഓരോന്നിന്റെയും ഉപയോഗം.

Test the fastness properties of dyed material 4 Understanding Water-Insoluble dyes: Sulphur Dyes - Method of application on cotton - Defects of sulphur dyeing and remedies. Vat Dyes - Types - Vatting - Method of application on Cotton/ Viscose material Ingrain Dyes: Azoic Colours - Dyeing - Diazotisation - Development of Colour on Cotton Brief idea on Application of Oxidised dyes, Mineral dyes and Natural Indigo on Cotton. Fastness properties of dyed material Text /Reference: T/R BookTitle/Author V.A. Shenai -Technology of Textile Processing (Volume 1 to 5) Mahajan T1 publishers Ahmedabad. R2 S.P.Mishra- Fibre science and technology R S Prayag - Scouring, Bleaching and Mercerizing and Dyeing of cellulose R3 fibres R4 E.R Trotman - Bleaching, Dyeing and Chemical Technology of Textile fibres R5 Srivastava - Recent process in Textile Bleaching and Dyeing R6 R RChakravarthi - A Glimpse of the chemical technology of Textile fibres R7 Nalankilli.G., Chemical Preparatory Process, NCUTE Publication, New Delhi Online resources: Sl.No Website Link

[http://textInfo.wordpress.com/2011/10/24/classification-of-textile-fibers- 1 according/](http://textInfo.wordpress.com/2011/10/24/classification-of-textile-fibers-1-according/) (accessed on 20 October 2013) 2

<https://nptel.ac.in/courses/116/102/116102026/> 3

<https://nptel.ac.in/courses/116/102/116102016/> 4 <https://swayam.gov.in/> 5

<http://eacharya.inflibnet.ac.in/vidya-mitra/>

Test the fastness properties of dyed material 4 Understanding Water-Insoluble dyes: Sulphur Dyes - Method of application on cotton - Defects of sulphur dyeing and remedies. Vat Dyes - Types - Vatting - Method of application on Cotton/ Viscose material Ingrain Dyes: Azoic Colours - Dyeing - Diazotisation - Development of Colour on Cotton Brief idea on Application of Oxidised dyes, Mineral dyes and Natural Indigo on Cotton. Fastness properties of dyed material Text /Reference: T/R BookTitle/Author V.A. Shenai -Technology of Textile Processing (Volume 1 to 5) Mahajan T1 publishers Ahmedabad. R2 S.P.Mishra- Fibre science and technology R S Prayag - Scouring, Bleaching and Mercerizing and Dyeing of cellulose R3 fibres R4 E.R Trotman - Bleaching, Dyeing and Chemical Technology of Textile fibres R5 Srivastava - Recent process in Textile Bleaching and Dyeing R6 R RChakravarthi - A Glimpse of the chemical technology of Textile fibres R7 Nalankilli.G., Chemical Preparatory Process, NCUTE Publication, New Delhi Online resources: Sl.No Website Link

[http://textInfo.wordpress.com/2011/10/24/classification-of-textile-fibers- 1 according/](http://textInfo.wordpress.com/2011/10/24/classification-of-textile-fibers-1-according/) (accessed on 20 October 2013) 2

<https://nptel.ac.in/courses/116/102/116102026/> 3

<https://nptel.ac.in/courses/116/102/116102016/> 4 <https://swayam.gov.in/> 5

<http://eacharya.inflibnet.ac.in/vidya-mitra/> is an official topic in Module 4 of

Textile Wet Processing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Test the fastness properties of dyed material 4 Understanding Water-Insoluble dyes: Sulphur Dyes - Method of application on cotton - Defects of sulphur dying and remedies. Vat Dyes - Types - Vatting - Method of application on Cotton/ Viscose material Ingrain Dyes: Azoic Colours - Dyeing - Diazotisation - Development of Colour on Cotton Brief idea on Application of Oxidised dyes, Mineral dyes and Natural Indigo on Cotton. Fastness properties of dyed material Text /Reference: T/R BookTitle/Author V.A. Shenai -Technology of Textile Processing (Volume 1 to 5) Mahajan T1 publishers Ahmedabad. R2 S.P.Mishra- Fibre science and technology R S Prayag - Scouring, Bleaching and Mercerizing and Dyeing of cellulose R3 fibres R4 E.R Trotman - Bleaching, Dyeing and Chemical Technology of Textile fibres R5 Srivastava - Recent process in Textile Bleaching and Dyeing R6 R RChakravarthi - A Glimpse of the chemical technology of Textile fibres R7 Nalankilli.G., Chemical Preparatory Process, NCUTE Publication, New Delhi Online resources: Sl.No Website Link
<http://textInfo.wordpress.com/2011/10/24/classification-of-textile-fibers-1> according/ (accessed on 20 October 2013) 2
<https://nptel.ac.in/courses/116/102/116102026/> 3
<https://nptel.ac.in/courses/116/102/116102016/> 4 <https://swayam.gov.in/> 5
<http://eacharya.inflibnet.ac.in/vidya-mitra/> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, test the fastness properties of dyed material 4 understanding water-insoluble dyes: sulphur dyes - method of application on cotton - defects of sulphur dying and remedies. vat dyes - types - vatting - method of application on cotton/ viscose material ingrain dyes: azoic colours - dyeing - diazotisation - development of colour on cotton brief idea on application of oxidised dyes, mineral dyes and natural indigo on cotton. fastness properties of dyed material text /reference: t/r booktitle/author v.a. shenai -technology of textile processing (volume 1 to 5) mahajan

t1 publishers ahmedabad. r2 s.p.mishra- fibre science and technology r s prayag - scouring, bleaching and mercerizing and dyeing of cellulose r3 fibres r4 e.r trotman - bleaching, dyeing and chemical technology of textile fibres r5 srivastava - recent process in textile bleaching and dyeing r6 r rchakravarthi - a glimpse of the chemical technology of textile fibres r7 nalankilli.g., chemical preparatory process, ncute publication, new delhi online resources: sl.no website link
[http://textInfo.wordpress.com/2011/10/24/classification-of-textile-fibers- 1](http://textInfo.wordpress.com/2011/10/24/classification-of-textile-fibers-1) according/ (accessed on 20 october 2013) 2 <https://nptel.ac.in/courses/116/102/116102026/> 3 <https://nptel.ac.in/courses/116/102/116102016/> 4 <https://swayam.gov.in/> 5 <http://eacharya.inflibnet.ac.in/vidya-mitra/> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Test the fastness properties of dyed material 4 Understanding Water-Insoluble dyes: Sulphur Dyes - Method of application on cotton - Defects of sulphur dyeing and remedies. Vat Dyes - Types - Vatting - Method of application on Cotton/ Viscose material Ingrain Dyes: Azoic Colours - Dyeing - Diazotisation - Development of Colour on Cotton Brief idea on Application of Oxidised dyes, Mineral dyes and Natural Indigo on Cotton. Fastness properties of dyed material Text /Reference: T/R BookTitle/Author V.A. Shenai -Technology of Textile Processing (Volume 1 to 5) Mahajan T1 publishers Ahmedabad. R2 S.P.Mishra- Fibre science and technology R S Prayag - Scouring, Bleaching and Mercerizing and Dyeing of cellulose R3 fibres R4 E.R Trotman - Bleaching, Dyeing and Chemical Technology of Textile fibres R5 Srivastava - Recent process in Textile Bleaching and Dyeing R6 R RChakravarthi - A Glimpse of the chemical technology of Textile fibres R7 Nalankilli.G., Chemical Preparatory Process, NCUTE Publication, New Delhi Online resources: Sl.No Website Link http://textInfo.wordpress.com/2011/10/24/classification-of-textile-fibers- 1 according/ (accessed on 20 October 2013) 2 https://nptel.ac.in/courses/116/102/116102026/ 3 https://nptel.ac.in/courses/116/102/116102016/ 4 https://swayam.gov.in/ 5 http://eacharya.inflibnet.ac.in/vidya-mitra/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Test the fastness properties of dyed material 4 Understanding Water-Insoluble dyes: Sulphur Dyes - Method of application on cotton - Defects of sulphur dyeing and remedies. Vat Dyes - Types - Vatting - Method of application on Cotton/ Viscose material Ingrain Dyes: Azoic Colours - Dyeing - Diazotisation - Development of Colour on Cotton Brief idea on Application of Oxidised dyes, Mineral dyes and Natural Indigo on Cotton. Fastness properties of

dyed material Text /Reference: T/R BookTitle/Author V.A. Shenai -Technology of Textile Processing (Volume 1 to 5) Mahajan T1 publishers Ahmedabad. R2 S.P.Mishra- Fibre science and technology R S Prayag – Scouring, Bleaching and Mercerizing and Dyeing of cellulose R3 fibres R4 E.R Trotman - Bleaching, Dyeing and Chemical Technology of Textile fibres R5 Srivastava – Recent process in Textile Bleaching and Dyeing R6 R RChakravarthi – A Glimpse of the chemical technology of Textile fibres R7 Nalankilli.G., Chemical Preparatory Process, NCUTE Publication, New Delhi Online resources: Sl.No Website Link
<http://textInfo.wordpress.com/2011/10/24/classification-of-textile-fibers-1> according/ (accessed on 20 October 2013) 2
<https://nptel.ac.in/courses/116/102/116102026/> 3
<https://nptel.ac.in/courses/116/102/116102016/> 4 <https://swayam.gov.in/> 5
<http://eacharya.inflibnet.ac.in/vidya-mitra/> **Textile Fibres** definition/meaning **Textile Fibres**. **Textile Fibres** **Textile Fibres**, steps, application **Textile Fibres** **Textile Fibres** **Textile Fibres**. Technical terms English-**Textile Fibres** **Textile Fibres**.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Execute cotton/ viscose fibre dyeing with water insoluble dyes.

	Execute the process of dyeing cotton and	
M4.01		4
Remembering	viscose materials with Water-insoluble dyes	
	Application of Ingrain dyes on textile	
M4.02		4
Understanding	materials	
M4.03	Test the fastness properties of dyed material	4
Understanding		
	Series Test - II	1
Contents:		
	Water-Insoluble dyes: Sulphur Dyes - Method of application on cotton - Defects of sulphur dying and remedies.	
	Vat Dyes - Types - Vatting - Method of application on Cotton/ Viscose material	
	Ingrain Dyes: Azoic Colours - Dyeing – Diazotisation - Development of Colour on Cotton	
	Brief idea on Application of Oxidised dyes, Mineral dyes and Natural Indigo on Cotton.	
	Fastness properties of dyed material	
Text /Reference:		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain know various chemicals used in textile wet 4 Remembering processing Discuss the preparation of cotton/ Viscose for. (3 marks)

Answer: Begin with the standard definition or identity of *know various chemicals used in textile wet 4 Remembering processing Discuss the preparation of cotton/ Viscose for*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 4 Understanding dyeing. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding dyeing*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Differentiate Singeing, Desizing & Scouring 3 Understanding Chemistry of Wetting: Definition of surface tension and contact angle - Relationship of surface tension and contact angle with wetting of textile materials Preparation of Cotton/ Viscose for Dyeing: Sequence of operations in wet processing of Cotton/ Viscose material. Singeing: Objectives - Methods - Hot plate, Roller, Gas singeing machines. De-sizing: Objectives - Methods - Rot steeping - Acid steeping, Enzymatic de-sizing. Scouring: Objectives and importance - mechanism of scouring- Types of Kiers. Features of a modern Kiers and its working. (3 marks)

Answer: Begin with the standard definition or identity of *Differentiate Singeing, Desizing & Scouring 3 Understanding Chemistry of Wetting: Definition of surface tension and contact angle - Relationship of surface tension and contact angle with wetting of textile materials Preparation of Cotton/ Viscose for Dyeing: Sequence of operations in wet processing of Cotton/ Viscose material. Singeing: Objectives - Methods - Hot plate, Roller, Gas singeing machines. De-sizing: Objectives - Methods - Rot steeping - Acid steeping, Enzymatic de-sizing. Scouring: Objectives and importance - mechanism of scouring- Types of Kiers. Features of a modern Kiers and its working*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 2

Define or explain 2 Understanding materials Use of continuous method of scouring and Bleaching process & compare hypochlorite 4 Understanding M2.02 and peroxide bleaching Gain the knowledge on the process and. (3 marks)

Answer: Begin with the standard definition or identity of 2 *Understanding materials Use of continuous method of scouring and Bleaching process & compare hypochlorite 4 Understanding M2.02 and peroxide bleaching Gain the knowledge on the process and.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain importance of Mercerisation to implement the 4 Understanding same in new situation. Bleaching: Objectives, Types of bleaching agents-Comparison of sodium and calcium hypochlorite's in terms of their properties, chemical behaviour and application - Method of bleaching with Sodium hypochlorite. Properties of hydrogen peroxide- Strength of hydrogen peroxide in volume & percentage - Bleaching with Hydrogen peroxide (Batch process).Continuous scouring and bleaching by using 'J' Box. Compare hypochlorite and peroxide bleaching. Mercerisation: Objects of mercerizing. Changes in the appearance of cotton by mercerizing - Various factors affecting the process of Mercerisation. Working of a Hank Mercerizing machine. Working of continuous Cloth Mercerizing machine Define BAN. Implement the process of dyeing Cotton/Viscose fibre using water-soluble. (3 marks)

Answer: Begin with the standard definition or identity of *importance of Mercerisation to implement the 4 Understanding same in new situation. Bleaching: Objectives, Types of bleaching agents-Comparison of sodium and calcium hypochlorite's in terms of their properties, chemical behaviour and application - Method of bleaching with Sodium hypochlorite. Properties of hydrogen peroxide- Strength of hydrogen peroxide in volume & percentage - Bleaching with Hydrogen peroxide (Batch process).Continuous scouring and bleaching by using 'J' Box. Compare hypochlorite and peroxide bleaching. Mercerisation: Objects of mercerizing. Changes in the appearance of cotton by mercerizing - Various factors affecting the process of Mercerisation. Working of a Hank Mercerizing machine. Working of continuous Cloth Mercerizing machine Define BAN.*

Implement the process of dyeing Cotton/Viscose fibre using water-soluble. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain 2 Understanding application Execute the application of Direct dyes,. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding application Execute the application of Direct dyes,.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding Reactive dyes on Cotton/ Viscose fibres Execute the application of Basic dyes on. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding Reactive dyes on Cotton/ Viscose fibres Execute the application of Basic dyes on.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding Cotton material. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding Cotton material.* State the governing principle, list the key components or stages, and close

with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Test the fastness properties of dyed material 2

Understanding Introduction to Dyeing: Define dyes - selection of dye - Classification of dyes based on the method of application - Terminologies related to dyes and dyeing - Material Liquor ratio, Shade Percentage, Percentage of Exhaustion, affinity, substantivity. General mechanism of dyeing. Important Natural dyes. Direct Dyes: Method of application of Direct Dye on Cotton and Viscose. After - treatments given to Direct dyed materials. Reactive Dyes: Properties - Types - method of application of Reactive Dyes on Cotton & Viscose Method of application of Basic dyes on cotton. Fastness properties of dyed material. (3 marks)

Answer: Begin with the standard definition or identity of *Test the fastness properties of dyed material 2* *Understanding Introduction to Dyeing: Define dyes - selection of dye - Classification of dyes based on the method of application - Terminologies related to dyes and dyeing - Material Liquor ratio, Shade Percentage, Percentage of Exhaustion, affinity, substantivity. General mechanism of dyeing. Important Natural dyes. Direct Dyes: Method of application of Direct Dye on Cotton and Viscose. After - treatments given to Direct dyed materials. Reactive Dyes: Properties - Types - method of application of Reactive Dyes on Cotton & Viscose Method of application of Basic dyes on cotton. Fastness properties of dyed material.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 4

Define or explain 4 Remembering viscose materials with Water-insoluble dyes Application of Ingrain dyes on textile. (3 marks)

Answer: Begin with the standard definition or identity of *4 Remembering viscose materials with Water-insoluble dyes Application of Ingrain dyes on textile.* State the

governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Understanding materials. (3 marks)

Answer: Begin with the standard definition or identity of 4 *Understanding materials*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Test the fastness properties of dyed material 4
**Understanding Water-Insoluble dyes: Sulphur Dyes - Method of application on cotton - Defects of sulphur dyeing and remedies. Vat Dyes - Types - Vatting - Method of application on Cotton/ Viscose material Ingrain Dyes: Azoic Colours - Dyeing - Diazotisation - Development of Colour on Cotton Brief idea on Application of Oxidised dyes, Mineral dyes and Natural Indigo on Cotton. Fastness properties of dyed material Text /Reference: T/R BookTitle/Author V.A. Shenai -Technology of Textile Processing (Volume 1 to 5) Mahajan T1 publishers Ahmedabad. R2 S.P.Mishra- Fibre science and technology R S Prayag - Scouring, Bleaching and Mercerizing and Dyeing of cellulose R3 fibres R4 E.R Trotman - Bleaching, Dyeing and Chemical Technology of Textile fibres R5 Srivastava - Recent process in Textile Bleaching and Dyeing R6 R RChakravarthi - A Glimpse of the chemical technology of Textile fibres R7 Nalankilli.G., Chemical Preparatory Process, NCUTE Publication, New Delhi Online resources: Sl.No Website Link
[http://textInfo.wordpress.com/2011/10/24/classification-of-textile-fibers- 1](http://textInfo.wordpress.com/2011/10/24/classification-of-textile-fibers-1) according/ (accessed on 20 October 2013) 2
<https://nptel.ac.in/courses/116/102/116102026/> 3
<https://nptel.ac.in/courses/116/102/116102016/> 4 <https://swayam.gov.in/> 5
[http://eacharya.inflibnet.ac.in/vidya-mitra/.](http://eacharya.inflibnet.ac.in/vidya-mitra/) (3 marks)**

Answer: Begin with the standard definition or identity of *Test the fastness properties of dyed material* 4 *Understanding Water-Insoluble dyes: Sulphur Dyes - Method of application on cotton - Defects of sulphur dyeing and remedies. Vat Dyes - Types - Vatting - Method of application on Cotton/ Viscose material Ingrain Dyes: Azoic Colours - Dyeing - Diazotisation - Development of Colour on Cotton Brief idea on Application of Oxidised dyes, Mineral dyes and Natural Indigo on Cotton. Fastness properties of dyed material* Text /Reference: T/R BookTitle/Author V.A. Shenai -Technology of Textile Processing (Volume 1 to 5) Mahajan T1 publishers Ahmedabad. R2 S.P.Mishra- Fibre science and technology R S Prayag – Scouring, Bleaching and Mercerizing and Dyeing of cellulose R3 fibres R4 E.R Trotman - Bleaching, Dyeing and Chemical Technology of Textile fibres R5 Srivastava – Recent process in Textile Bleaching and Dyeing R6 R RChakravarthi – A Glimpse of the chemical technology of Textile fibres R7 Nalankilli.G., Chemical Preparatory Process, NCUTE Publication, New Delhi Online resources: Sl.No Website Link
[http://textInfo.wordpress.com/2011/10/24/classification-of-textile-fibers- 1](http://textInfo.wordpress.com/2011/10/24/classification-of-textile-fibers-1) according/ (accessed on 20 October 2013) 2 <https://nptel.ac.in/courses/116/102/116102026/> 3 <https://nptel.ac.in/courses/116/102/116102016/> 4 <https://swayam.gov.in/> 5 <http://eacharya.inflibnet.ac.in/vidya-mitra/>. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **know various chemicals used in textile wet 4 Remembering processing Discuss the preparation of cotton/ Viscose for.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **2 Understanding materials Use of continuous method of scouring and Bleaching process & compare hypochlorite 4 Understanding M2.02 and peroxide bleaching Gain the knowledge on the process and.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **2 Understanding application Execute the application of Direct dyes,.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **4 Remembering viscose materials with Water-insoluble dyes Application of Ingrain dyes on textile.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- [http://textInfo.wordpress.com/2011/10/24/classification-of-textile-fibers- according/](http://textInfo.wordpress.com/2011/10/24/classification-of-textile-fibers-according/) (accessed on 20 October 2013)
- <https://nptel.ac.in/courses/116/102/116102026/>
- <https://nptel.ac.in/courses/116/102/116102016/> <https://swayam.gov.in/>
<http://eacharya.inflibnet.ac.in/vidya-mitra/>

IN-PAGE SEARCH

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Prepared from the supplied official SITTTTR syllabus PDF for Course 3062. Verify institutional updates against the current official curriculum.